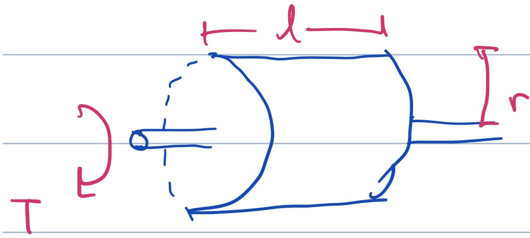
Tangential Stress:

$\sigma = k\ \bar{B} . \bar{A}$ N/m^2 , $\bar{B}$: magnetic load $\bar{A}$: electrical load

k: constant



$$\sum F = \sigma . 2\pi r l$$

$$T = \sigma . 2\pi r^2 l$$

$$= k . \bar{B} . \bar{A} . 2\pi r^2 l$$

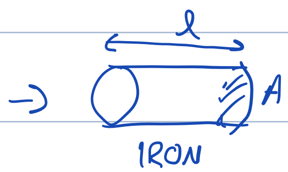
$T = k . \bar{B} . \bar{A} . \text{Volume}$: Torque and Volume proportional to each other.

This determines the size.

$$P = T . w$$

$$= k . \bar{B} . \bar{A} . \text{Volume} . w$$

Permability (μ H/m)

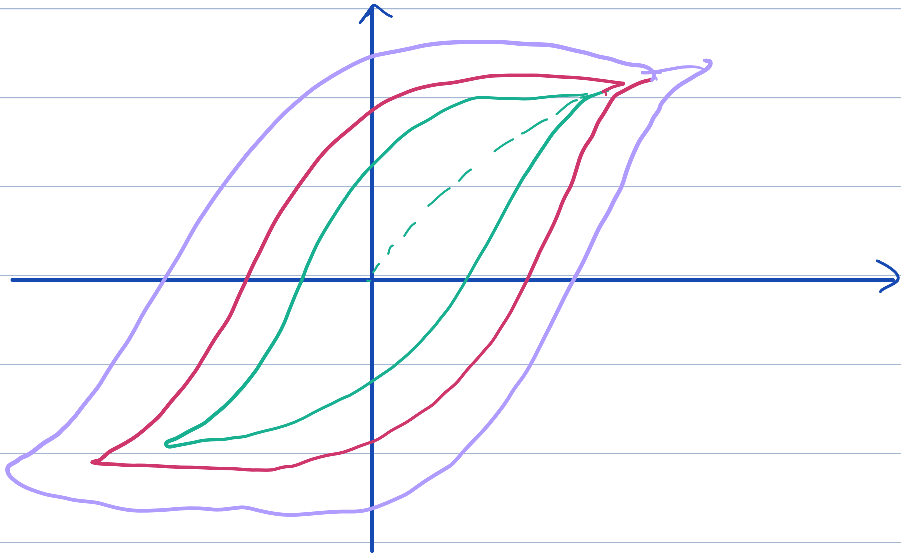


$R = \dfrac{l}{\mu . A}$, $L = \dfrac{N^2}{R}$ $\Rightarrow$ $R \rightarrow \dfrac{1}{H}$ unit

$\mu \rightarrow H/m$

H : Magnetic field density is independent of material properties.

Hysteresis Loss: $P_H \propto f$



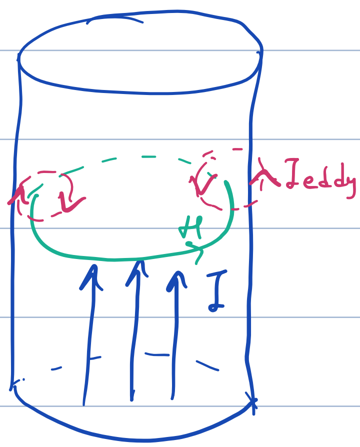
Eddy Current: $P_e \propto B^2, f^2$

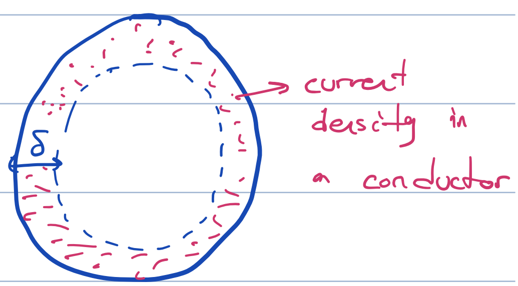
$\vec{J} = \sigma . \vec{E}$ in electric circuits $P_{loss} = V^2/R$

in magnetic $e = N \frac{d\Phi}{dt} \rightarrow \propto$ Bapplied freq. $\Rightarrow P_e \propto B^2, f^2$

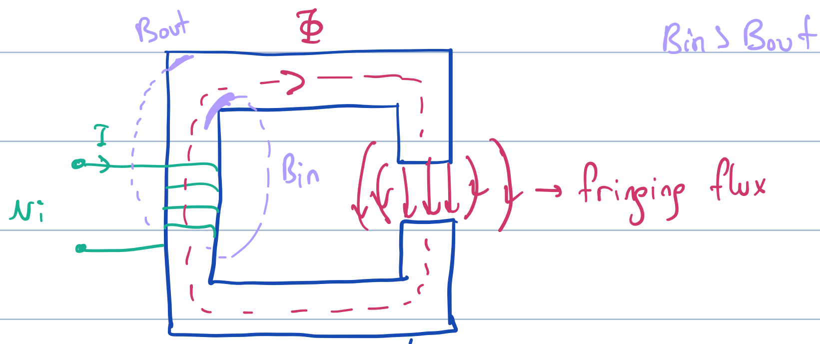
Skin Depth:

δ: Skin depth



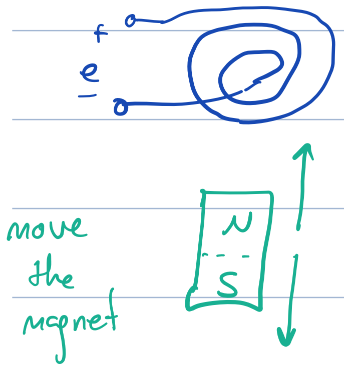


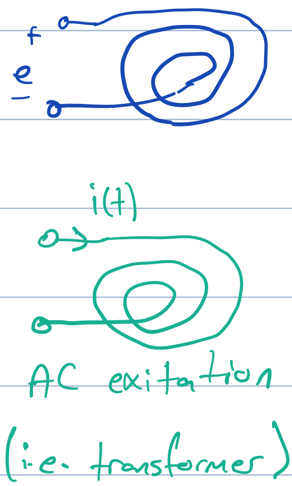
Assumptions:

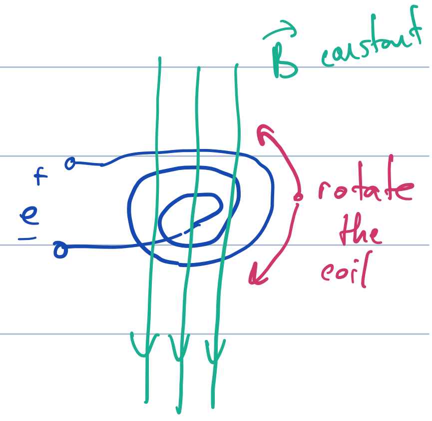


↳ μ is not ∞ and have saturation.

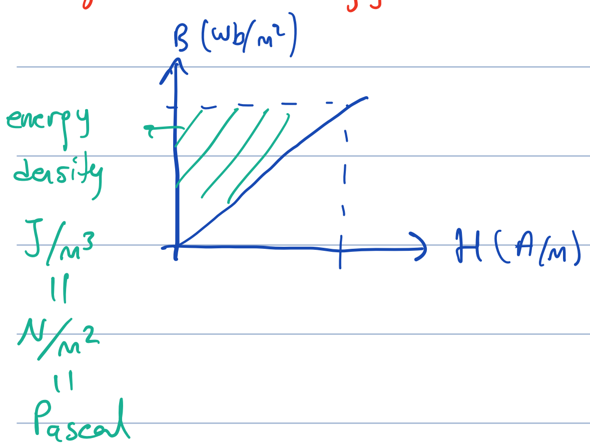
Faraday Law's of Induction:



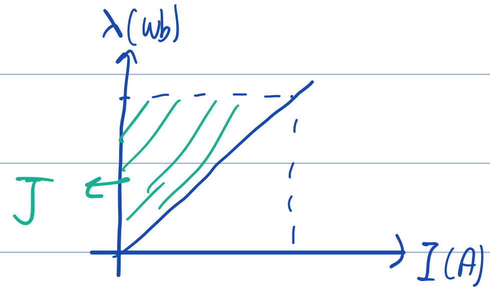




Magnetic Energy:

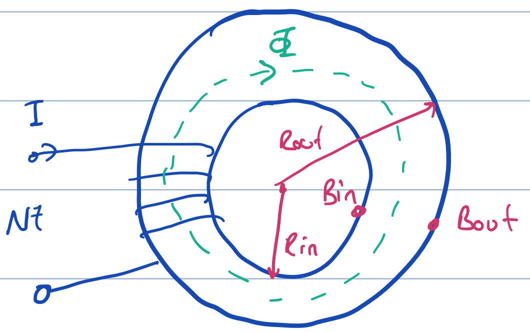


linear material



$$W = \int H \, dB \overset{\rightarrow \text{Area}}{=} \frac{BH}{2} = \frac{B^2}{2\mu}$$

Magnetic Core:

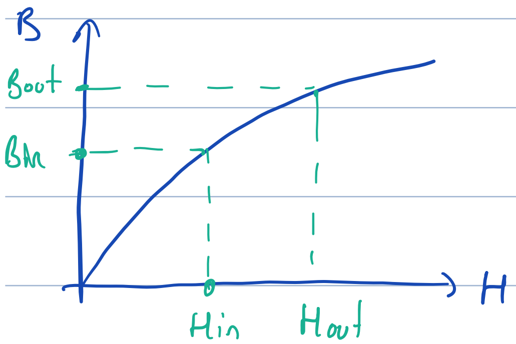


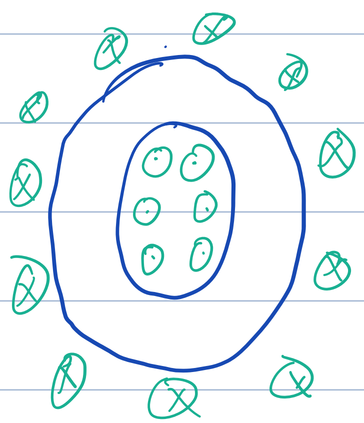
Relation of Bin – Bout

$$\oint H \cdot dl = N \cdot I$$

$$H \cdot 2\pi R_{in} = \underbrace{N \cdot I}_{\text{known}}$$

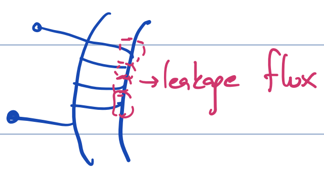
$\hookrightarrow B = H \cdot \mu \Rightarrow H = B/\mu$

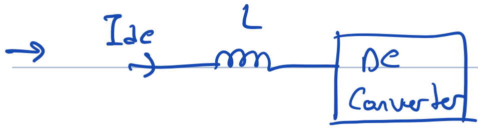


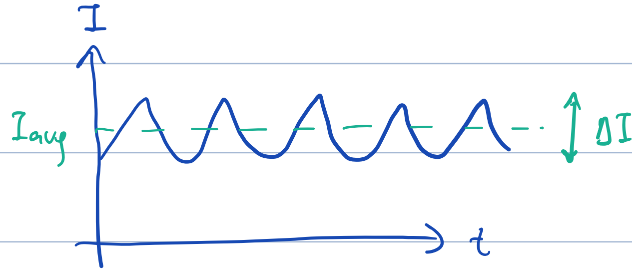


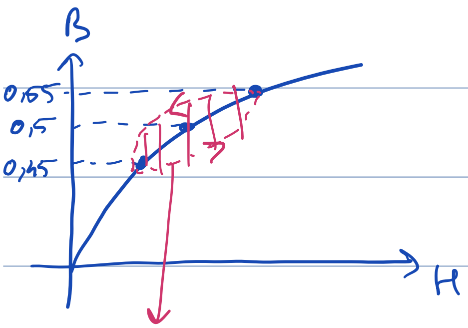
Problems:

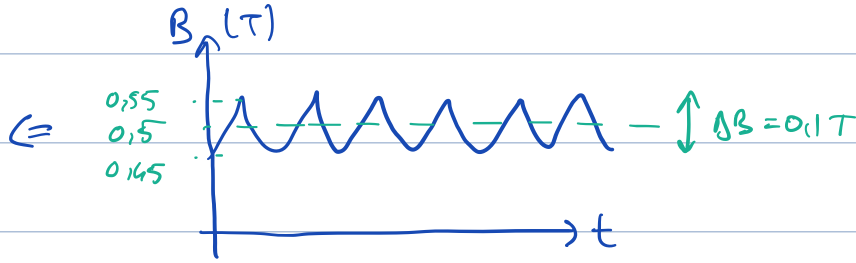
- Cooling is harder
- Copper losses may increase depend on the oper.
- If B is limited $\int \frac{B}{\mu} dl = NI$. So $N\nearrow$ $I\swarrow$
- $L\nearrow$







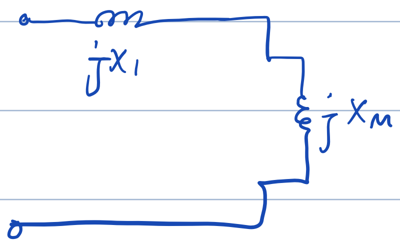


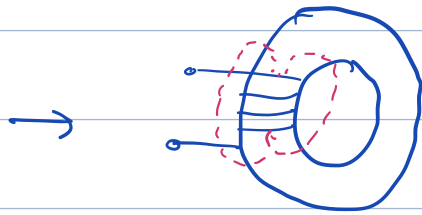


Hysteresis Losses

Equivalent Circuit of Transformer:

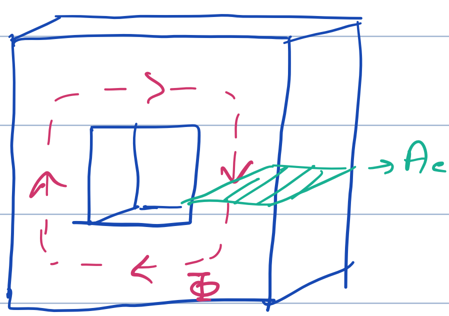
Assume





jX_1: linkage inductance (related w/ air)

jX_M: Core's inductance



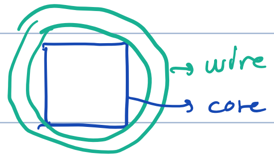
$$e = N \cdot \frac{d\Phi}{dt} = N \frac{d(\int BA)}{dt}$$

$$\Rightarrow e = N \cdot A \cdot \frac{2\pi f}{\sqrt{2}} \cdot B$$

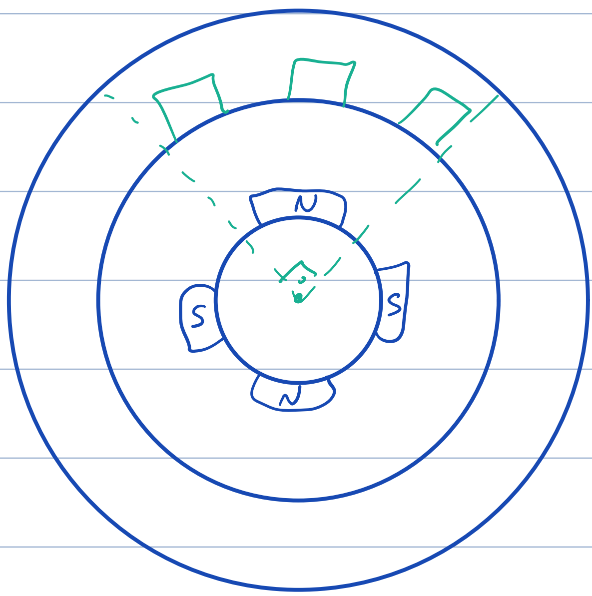
↳ rms

Design tips:

- At nominal voltage usually $P_{copper} > P_{core}$ but transformers work at their %50 usually. So $P_{core} \geq P_{copper}$ sometimes.
- Total cable area is not equal to the window area. Fill factor is an important parameter.
- In the calculation of R_1 & R_2, resistivity change with Temperature.
-



this is more effective.



Pole pitch: $\tau_p = \dfrac{\pi . D}{2p}$

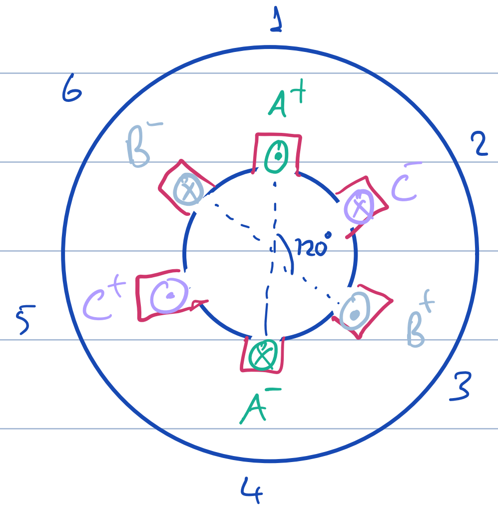
Slot " : $\tau_u = \dfrac{\pi D}{Q}$ Slot angle: $\alpha_u = \dfrac{2\pi}{(Q/p)}$ electrical angle

3 Phase Integral Slot Stator Winding:

$m = 3$: # of phase
$p = 1$: # of pole-pair
$q = 1$: slots per pole per phase

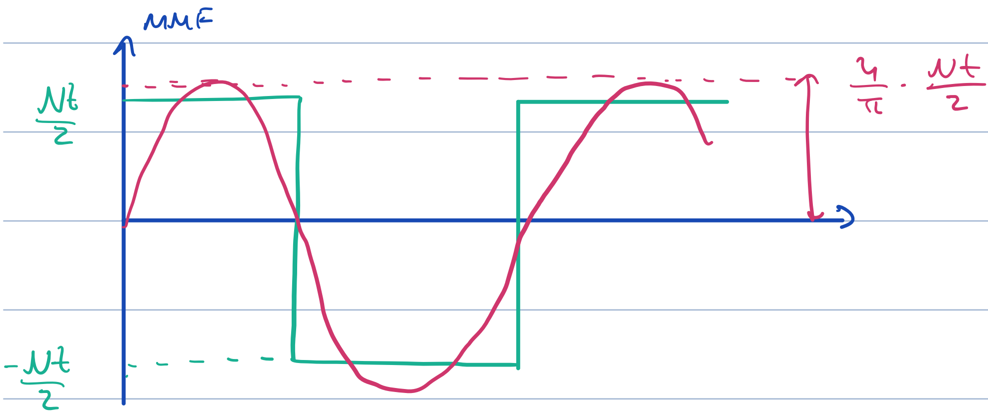
$Q = 6$: # of slots.

$q = Q/2pm$

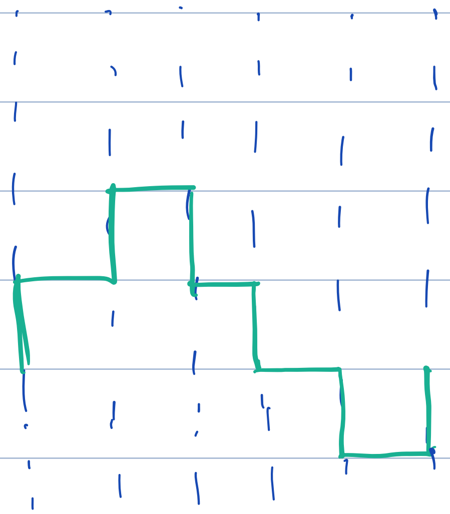


# of slots	1	2	3	4	5	6
phase	A^+	C^-	B^+	A^-	C^+	B^-

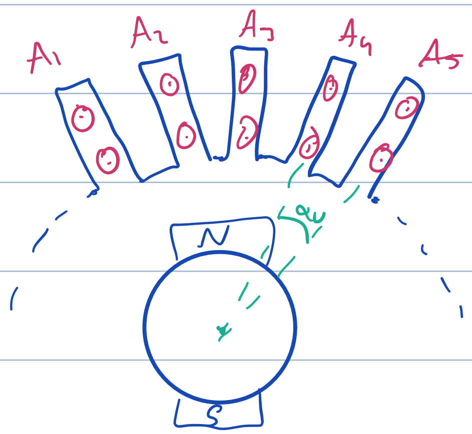
MMF from a single phase:



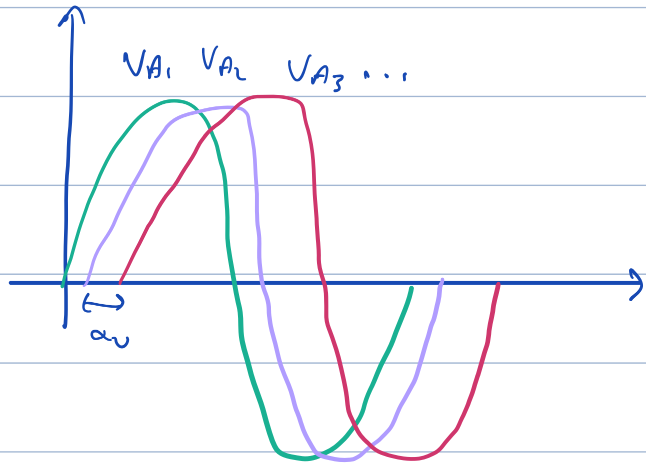
# of slots	1	2	3	4	5	6
phase	A^+	C^-	B^+	A^-	C^+	B^-



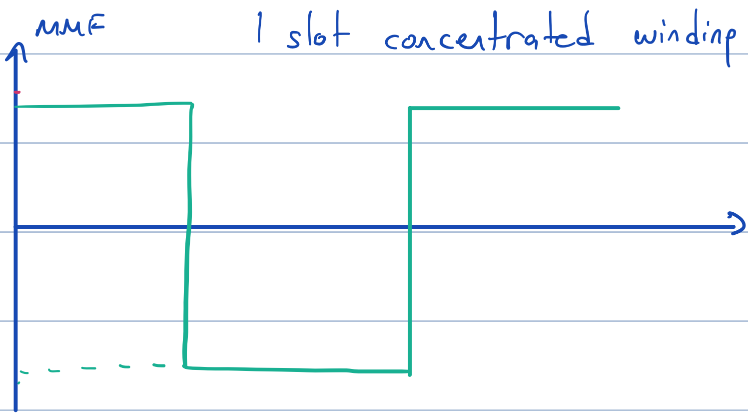
→ MMF from 3-phase

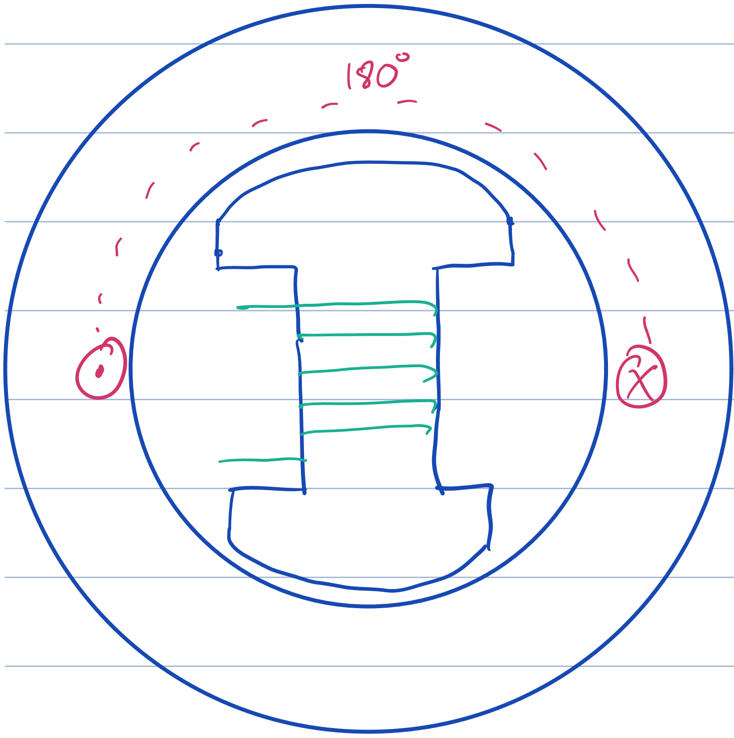


$$\alpha_u = \frac{2\pi}{30} = 12^\circ$$

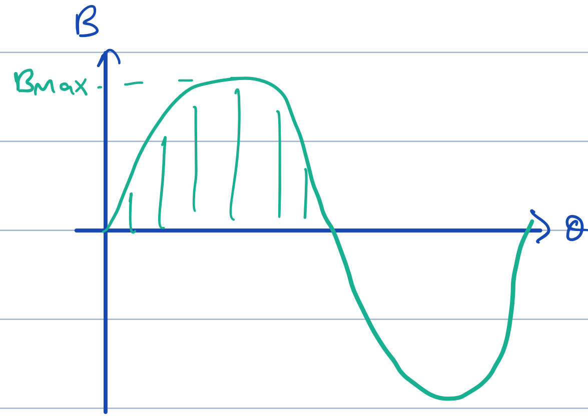


1 slot concentrated winding

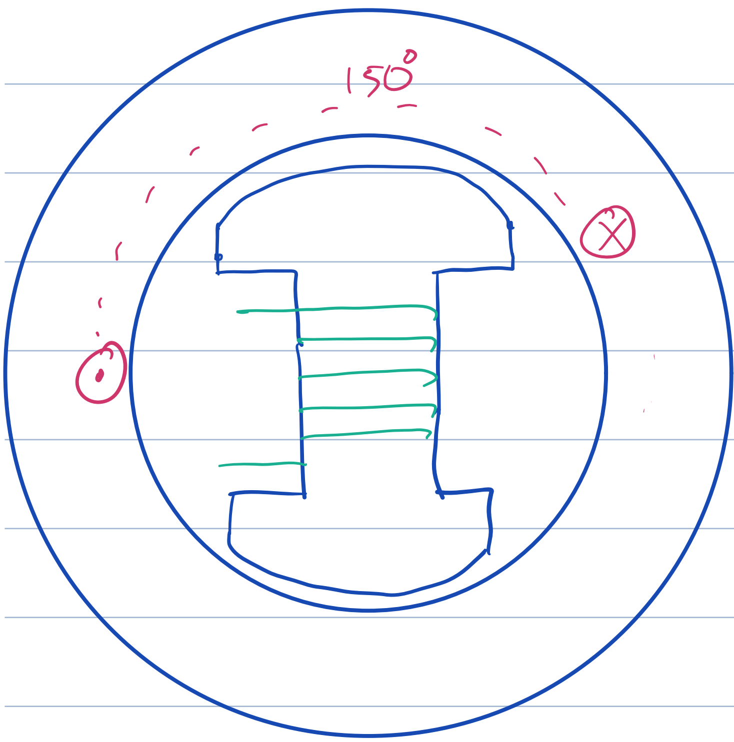




Full-pitch winding

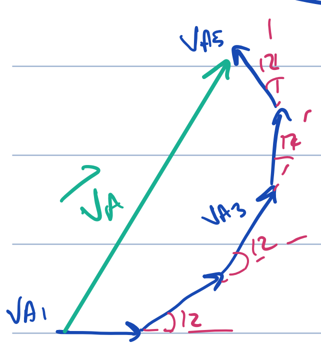


$$\Phi = \int_0^{\pi} B\,dA$$



5/6 pitched winding

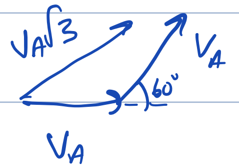
$$\Phi = \int_0^{\pi} B\,dA$$



$$k_d = \frac{\text{vector sum of voltages}}{\text{algebric} \quad ''}$$

→ Distribution factor

k_d of 2 winding seperated by $\pi/3$ is:

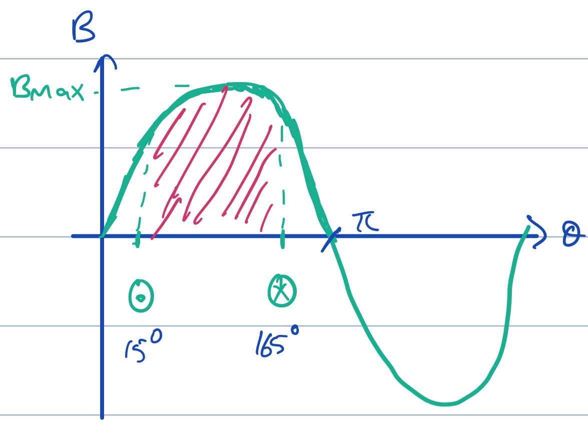


$$\Rightarrow k_d = \frac{V_A\sqrt{3}}{2V_A} = \frac{\sqrt{3}}{2}$$

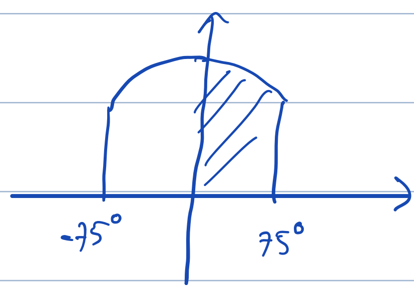
In general $\quad k_d = \dfrac{\sin\left(q\frac{\alpha}{2}\right)}{q.\sin\left(\frac{\alpha}{2}\right)}$

---

pitch factor: $\quad k_p = \sin\left(\frac{\lambda}{2}\right)$, λ: coil pitch in electrical angle



$\Rightarrow$



$\rightarrow$ Area : $2.\sin(75)$

$$k_p = \frac{2\sin(75)}{2}$$

$$\Phi = \int_0^{\pi} B\,dA$$

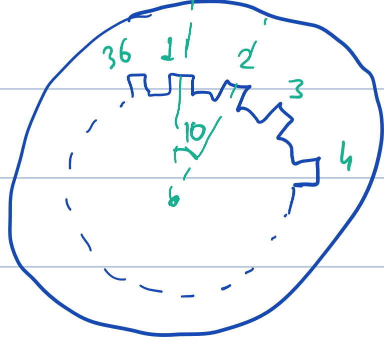
$k_p(n) = \sin\left(\frac{n\lambda}{2}\right)$, n is the harmonic number.

$$k_d(n) = \frac{\sin\left(qn\frac{\alpha}{2}\right)}{q.\sin\left(n\frac{\alpha}{2}\right)}$$

$k_w = k_p.k_d$, winding factor.

Ex 36 slots, 4 pole, 3 phase

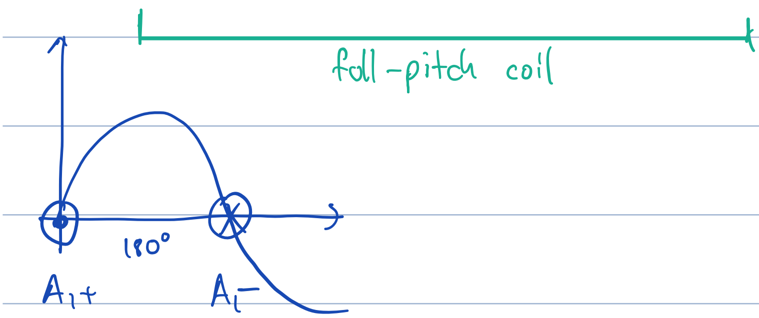
$q = \frac{36}{4 \cdot 3} = 3$

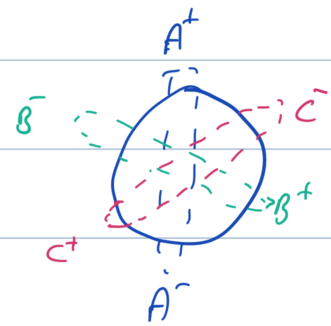


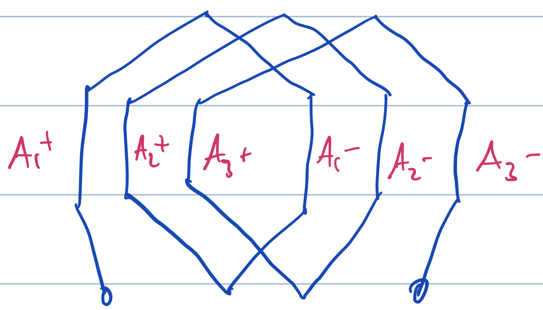
10° mechanical, 20° electrical.

if full pitch

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	A_1+	A_2+	A_3+	C_1-	C_2-	C_3-	B_1+	B_2+	B_3+	A_1-	A_2-	A_3-	C_1+	C_2+	C_3+	B_1-	B_2-	B_3-

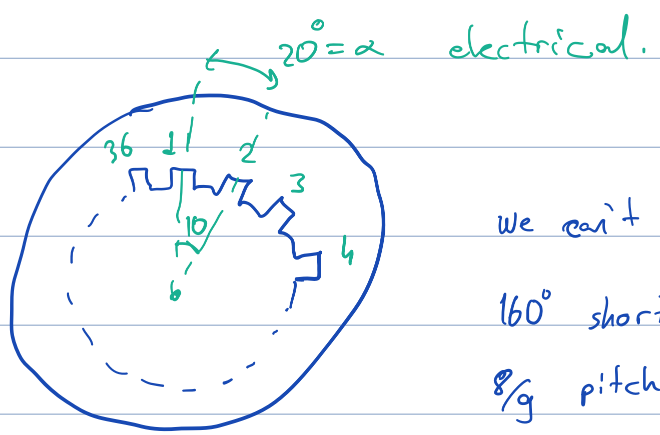






ph A

if short pitch



we can't choose 150° so

160° short-pitch. 20° steps.

8/9 pitched.

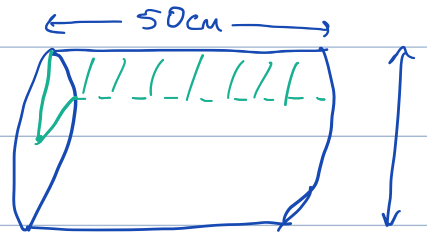
Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
B_6^-	A_1^+	A_2^+	A_3^+	C_4^-	C_5^-	C_6^-	B_1^+	B_2^+	B_3^+	A_4^-	A_5^-	A_6^-	C_1^+	C_2^+	C_3^+	B_4^-	B_5^-	B_6^-	A_1^+ ...
A_4^+	A_5^+	A_6^+	C_1^-	C_2^-	C_3^-	B_4^+	B_5^+	B_6^+	A_1^-	A_2^-	A_3^-	C_4^+	C_5^+	C_6^+	B_1^-	B_2^-	B_3^-	A_4^+	

Double-layer winding configuration

Ex: A rotating magnetic flux is created at 50 Hz, 3ph, 600 rpm.
Machine has
- 180 slots, double-layer winding, Y connected.
- Each coil has 3 turns with a span of 15 slots.
- Armature diameter: 135 cm
- Effective axial depth: 0,5 m

→ $N_s = \frac{120 f}{p} \Rightarrow p = 10$ poles, $N_s = 600$

$q = \frac{180}{3 \cdot 10} = 6$



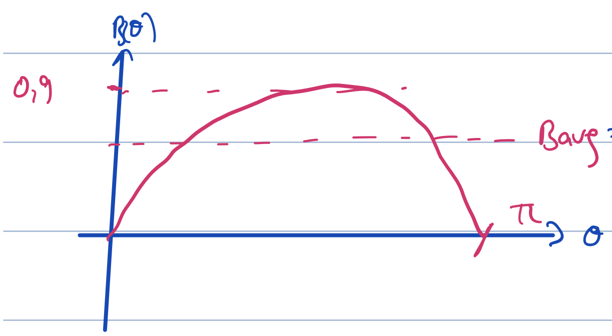
pole area:

$\frac{\pi \cdot D \cdot L}{2p} = \frac{\pi \cdot 1{,}35 \cdot 0{,}5}{10}$

pole-pairs

$= 0{,}212 \ m^2$

$B(\theta) = 0{,}9 \cdot \sin(\theta) + 0{,}25 \sin(3\theta) + 0{,}15 \sin(5\theta)$



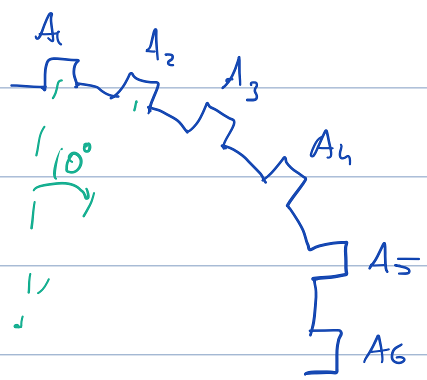
$B_{avg} = \frac{2}{\pi} \cdot \hat{B}$

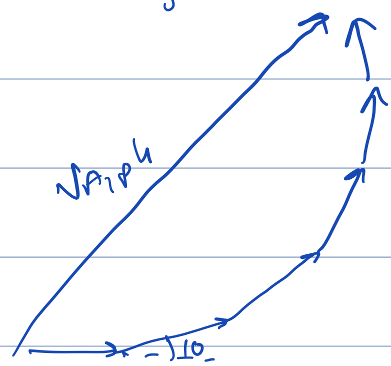
max flux $= \hat{\Phi} = \int_0^{\pi} B(\theta)\, d\theta$

$= B_{avg} \cdot$ pole-area

$\hat{\Phi} = \frac{2}{\pi} \cdot 0{,}9 \cdot 0{,}212 = 121{,}4 \text{ mWb}$

$q=6$, $\alpha = \frac{360}{180} \cdot \frac{10}{2} = 10^\circ$ in electrical degrees.





$k_{d_1} = \frac{\sin\left(q \frac{\alpha}{2}\right)}{q \cdot \sin\left(\frac{\alpha}{2}\right)} = 0{,}956$ $\quad k_{d_3} = \frac{\sin\left(q \cdot \frac{3\alpha}{2}\right)}{q \cdot \sin\left(\frac{3\alpha}{2}\right)} = 0{,}643$ $\quad , k_{d_5} = 0{,}197$

$\lambda = 150^\circ = (10 \cdot 15 \;, \alpha \cdot N_{span})$ $\quad \frac{5\pi}{6}$ pitched

$k_{p_1} = \sin\left(\frac{5\pi}{6 \cdot 2}\right) = 0{,}9659$ $\quad k_{p_2} = -0{,}707$, $k_{p_3} = 0{,}259$

180° phase shift

$k_{w_1} = 0{,}922$ $\quad k_{w_2} = -0{,}454$ $\quad k_{w_3} = 0{,}05$

$V_{ind.} = 4{,}44 \cdot f \cdot N \cdot B \cdot A \cdot k_w$

N: $q=6$, each coil has 3 turns. poles connected in series

60 slots/phase $\Rightarrow$ $6 \cdot 60 = 360$ conductor, 180 turn/phase $(\underset{q}{6} \cdot 3 \cdot \underset{\text{pole}}{10})$

B: $0{,}9 \cdot \frac{2}{\pi} = B_{avg}$

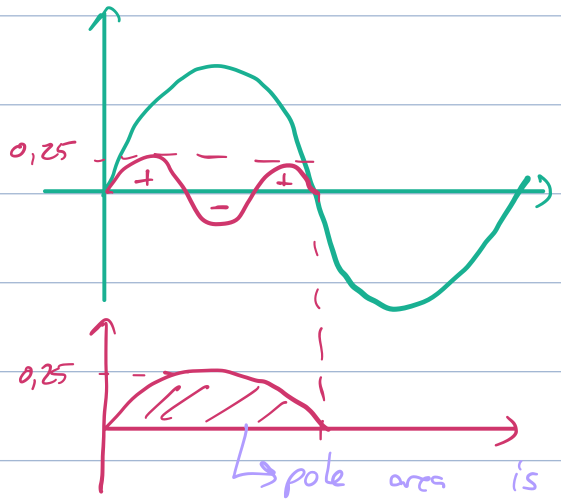
$A = 0{,}212 \text{ m}^2$

$\Rightarrow V_1 = 4480 \text{ V}$

$V_2 = 613 \text{ V}$

$V_3 = 49{,}5 \text{ V}$

V_2 için f 150 Hz oluyor ama efektif pole-area 3'te 1'ine düşüyor



↳ pole area is 1/3 of above.

Fractional Winding:

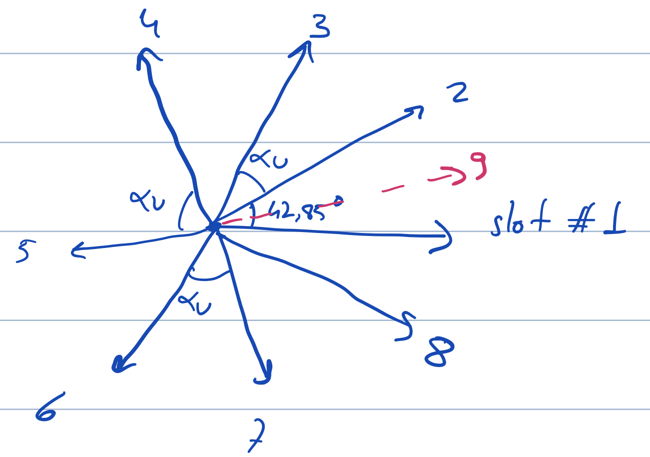
Assume a machine with 42 slots 10 pole.

$$q = \frac{42}{10 \cdot 3} = \frac{7}{5}$$

for stable machines, it is better that slot numbers is multiple of phase number.

phase difference between 2 coils/slots:

$$\alpha_v = 360 \cdot \left(\frac{10}{2}\right) \cdot \frac{1}{42} = 42{,}85^\circ$$



for a large induced emf group 1-8-9 or we can use 5 by reversing it. We can use return cable.

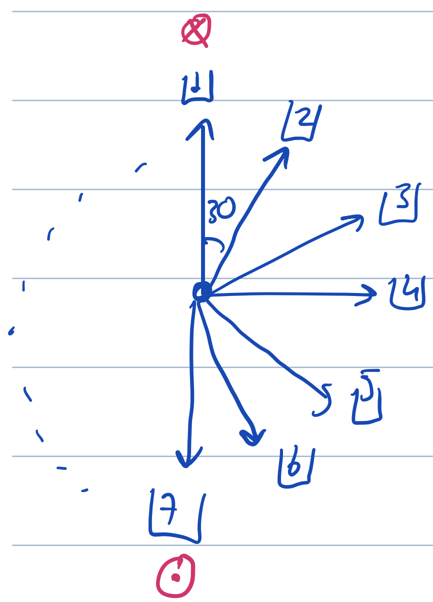
Coil pitch for 4:

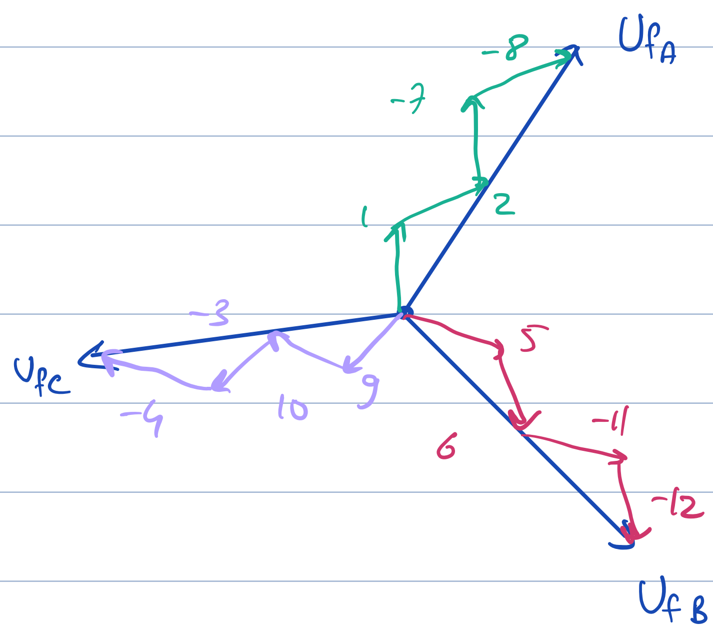
$42{,}85 \cdot 4 = 171{,}5 \Rightarrow$ coil pitch : $\sin(171{,}5/2) = 0{,}997$

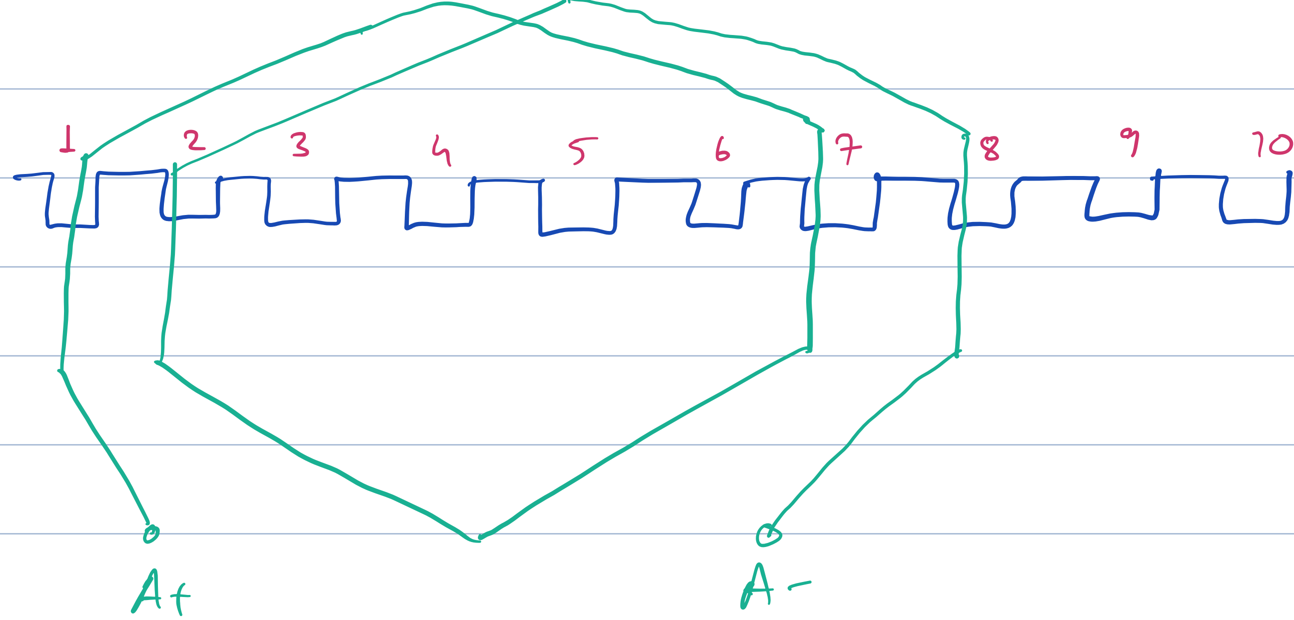
Winding Voltage Phasor Diagrams:

Assume a machine with 2 pole - 3 ph - 12 slots. ⇒ Integer Slot

$\alpha_u = 360 \cdot \left(\frac{2}{2}\right) \cdot \frac{1}{12} = 30°$, $q = \frac{12}{3 \cdot 2} = 2$





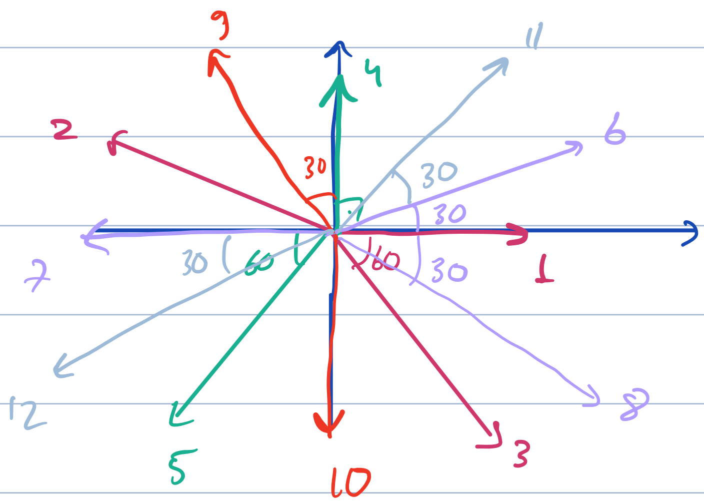


Ex

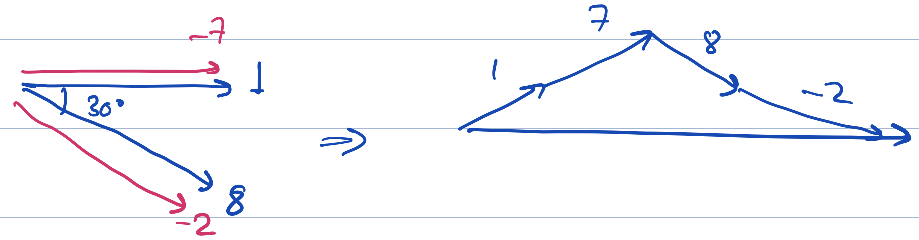
12 slots 10 pole single layer

$$\alpha_u = 360 \cdot \left(\frac{10}{2}\right) \cdot \frac{1}{12} = 150^\circ$$

$\Rightarrow$



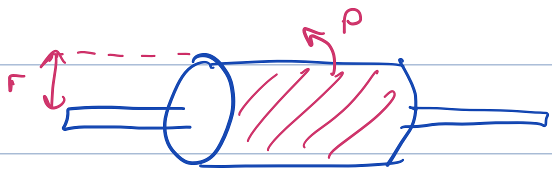
$\Rightarrow U_{f_A}$:



Maxwell Stress Tensor:

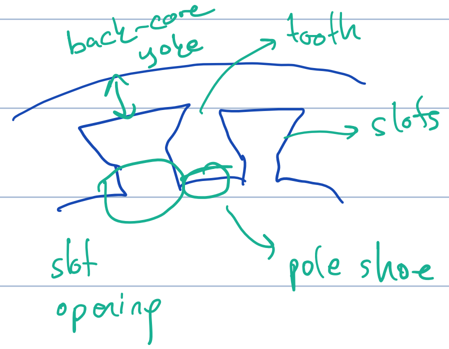
$$\sigma_F = \frac{1}{2} \cdot \mu_0 H^2 \quad \text{or} \quad \sigma_F = \frac{1}{2\mu_0} \cdot B^2 \qquad [Pa] \rightarrow \text{pascal} : N/m^2$$

Shear Stress:



$$N/m^2 \cdot \text{Surface } A \cdot r = T$$

Magnetic loading:



$$\bar{B} = \frac{p \cdot \Phi_{pp}}{\pi \cdot D_i L}$$

Electric Loading:

$$\bar{A} = \frac{N_{turn,slot} \cdot I \cdot Q}{\pi \cdot D_i}$$

Specific Machine Constant:

$$S = \underbrace{m}_{\#\text{of phases}} \cdot \underbrace{E_m}_{\text{induced voltage}} \cdot I_s \rightarrow \text{current}$$

$$A = \frac{2mN_s}{\pi D} \cdot I_s$$

$$S = m \cdot \frac{1}{\sqrt{2}} \, w \cdot N_s \cdot k_w \cdot \Phi_m \cdot I_s$$

$$= m \cdot \frac{1}{\sqrt{2}} \cdot w N_s \cdot k_w \cdot \Phi_M \cdot \frac{A \pi D}{2 m N_s}, \quad w = 2\pi p \, n_{syn}$$

electric loading; pole pair; mechanical syn. speed

$$= \frac{\pi^2}{\sqrt{2}} \cdot n_{syn} \cdot k_w \cdot A \cdot \hat{B} \cdot D^2 \cdot l$$

$$= C \cdot D^2 \cdot l \cdot n_{syn} \quad , \quad C = \frac{\pi^2}{2} \cdot k_w \cdot \hat{A} \cdot \hat{B}$$

Ex Pyrhonen 6.1

Diameter 98 mm   length 82 mm

4 kW, 50 Hz, 2 pole induction motor

Assume slip is 1% $s = 0{,}01$

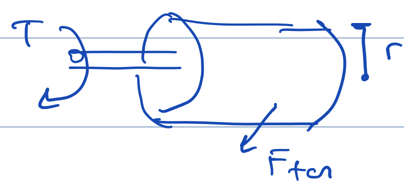
Find machine const. and avg. tangential stress

$$T = \frac{4\,kW}{99\pi} \rightarrow 2\pi f(1-s) = 99\pi \quad \Rightarrow T = 12{,}86 \text{ Nm}$$

Rotor Area :

$$S_r = \pi \cdot D_r \cdot l = \pi \cdot 0{,}098 \cdot 0{,}082 = 0{,}0252 \; m^2$$

$$F_{tan} = \frac{T}{r} = \frac{12{,}86}{0{,}049} = 262 \; N$$



$$\text{Stress} \quad \sigma = \frac{F_{tan}}{S_r} = \frac{262}{0{,}0252} = 10\,400 \; N/m^2$$

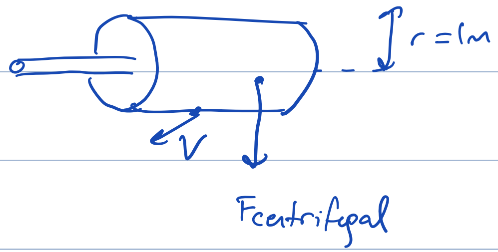
$$P_{mech} = C_{mech} \cdot D^2 \, l \, f_{syn} \Rightarrow C_{mech} = \frac{4000}{0{,}098^2 \cdot 0{,}082 \cdot 50} = 102 \; kWs/m^2$$

Aspect Ratio: $x = \frac{L'}{D}$ → axial length

Asynchronous $\quad x \cong \frac{\pi}{2pp} \cdot \sqrt[3]{pp} \quad$ pp: pole pair

Synchronous $x \cong \frac{\pi}{4pp}\sqrt{pp}$ → Kontrol et

Ex



$N_r = 20000$ rpm => $w = 2000$ rad/s

$V = w.r = 2000$ m/s too much!

$a = w^2.r = (2000)^2.r = 4.10^6$ m/s² => $F_{cent.} = ma = 40000\,g$

Ex

30kW, 690 V, 50 Hz, 4 pole, 3ph.

Find the main dimensions.

$C_{mech} = 150$ kW.s/m3

→ $x \cong \frac{L'}{D} = \frac{\pi}{2\underbrace{p}_{\text{pole pairs}}} \sqrt[3]{p} = 0{,}989 \approx 1$

$P_{mech} = C_{mech} . \underbrace{D^2 . l}_{\text{volume}: D^3} . n_{syn}$ → rotational speed

$D = \sqrt[3]{\frac{P_{mech}}{C_{mech} . n_{syn}}} = \sqrt{\frac{30}{150 . \underbrace{25}_{Hz : \frac{f}{pp}}}} \cong 200{,}7$ mm

find the tangential stress (σ) if speed is 1474 rpm

$$T = \frac{P}{\omega} = \frac{30000}{2\pi \cdot \frac{1474}{60}} = 194{,}4 \text{ Nm}$$

$$\sigma_{tan} = \frac{T}{S_{rot} \cdot r} = \frac{194{,}4}{\pi D \cdot l \cdot \left(\frac{D}{2}\right)} = 15{,}5 \text{ kPa}$$

Ex

30kW, 4 pole induction mach.

$D = 200$ mm, $L = 200$ mm, Stator circumference $= 628$ mm

Assume $q = 4 \Rightarrow N_{slot} = 4 \cdot 3 \cdot 4 = 48$

Tooth thickness $= 0{,}5 \cdot \frac{628}{48} = 6{,}5$

If $q=3$, Coil span $= 9$ slots, A $8/9$ or $7/9$ can be used

Suitable air gap:

$$\varepsilon = 1{,}6 \cdot 0{,}18 \cdot 0{,}006 \cdot (30k)^{0{,}4}$$

↳ %60 increase due to heavy duty motor

$$\cong 0{,}88 \text{ mm} = 0{,}9 \text{ mm}$$

Ex

Calculate the max diameter for smooth stell cylinder having a small bore. Speed is 15k rpm. Yield strength 300 MPa. Density is 7860 kg/m3.

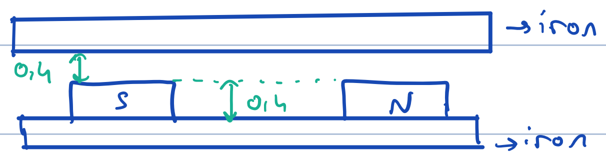
$\rightarrow$ for steel poisson ratio

$$c' = \frac{(3+v)}{4} = \frac{(3+0{,}29)}{4} = 0{,}823$$

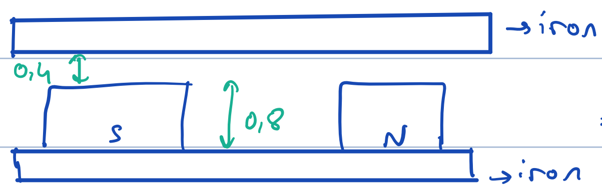
$\rightarrow$ Denklemi Kontrol Et

$$\sigma_{mech} = c' \cdot \rho \cdot r^2 \Omega^2 < 300 \text{ Mpa}.$$

$$\sigma_{max} = \sqrt{\frac{300\text{Mpa}}{0{,}823 \cdot 7860 \cdot \left(\frac{2\pi\, 15k}{60}\right)^2}} = 0{,}13 \text{ m}$$



$\Rightarrow B \cong 0{,}4\,T$



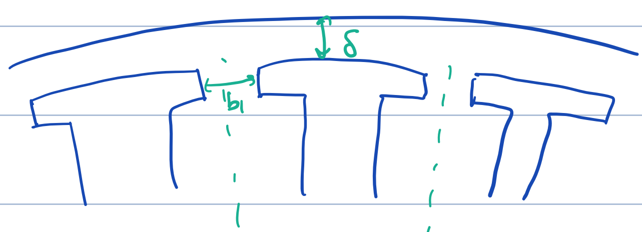
$\Rightarrow B' < 0{,}8\,T$

Because PM has $\mu_r \cong 1{,}05$, acts like an air gap

Ex (3.1)

air gap $\delta = 0{,}8$ mm

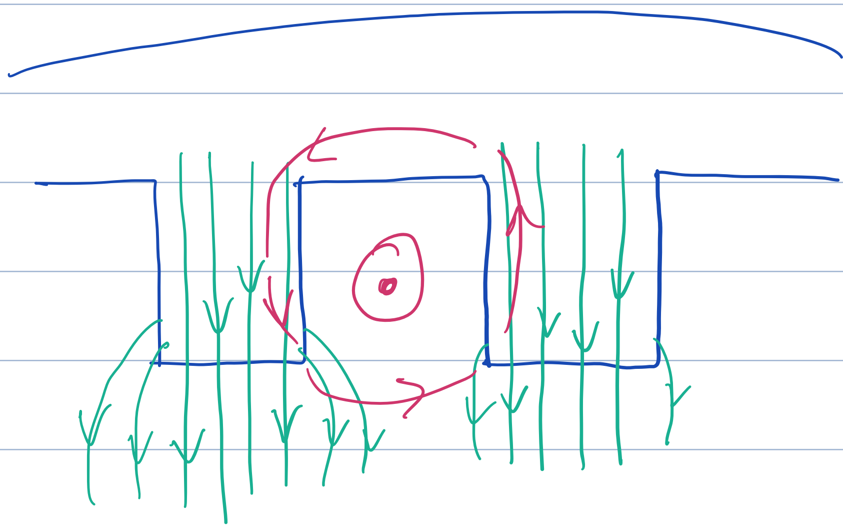
slot opening $b_1 = 3$ mm , stator slot pitch $\tau_u = 10$ mm

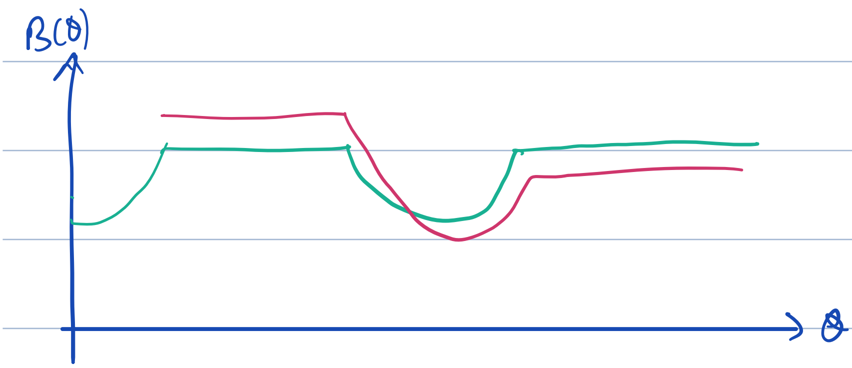


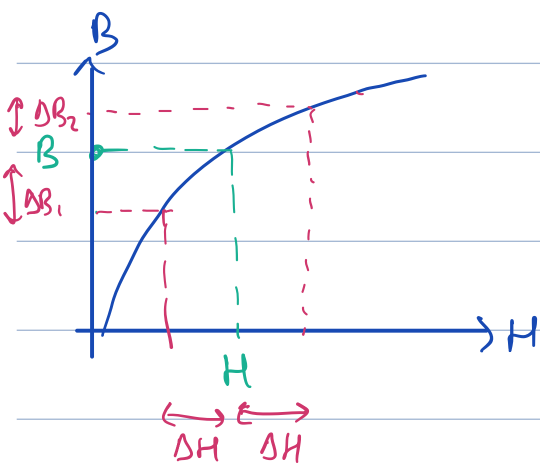
H6

$$k_c = \frac{\tau_0}{\tau_0 - \kappa b_1}, \quad \kappa = \frac{b_1/\delta}{5 + b_1/\delta} = 0{,}429, \quad \kappa b_1 \cong 1{,}3 \text{ mm}$$

$$k_c = \frac{10}{10 - 1{,}3} \cong 1{,}15 \Rightarrow \delta_e = k_c \cdot \delta = 0{,}918 \text{ mm} \quad \text{(effective air-gap)}$$







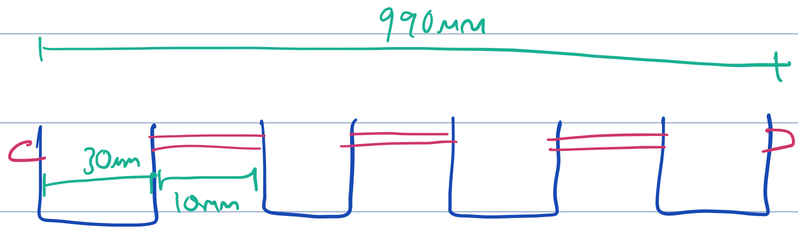
$\Delta B_1 \neq \Delta B_2$

Ex (3.2)

Stator core : 990 mm

25 sub-stacks of lamination, each 30mm long and 10mm cooling duct.

air gap is 3mm.



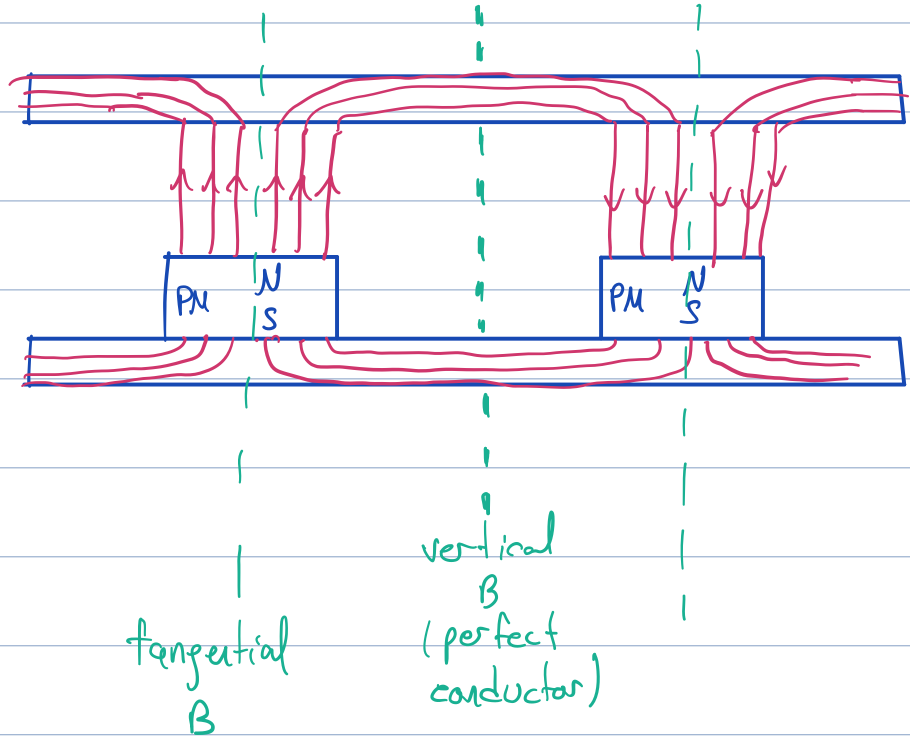
24 cooling ducts

25 stator sectors

$\Rightarrow 25 \cdot 30 + 24 \cdot 10 = 990\,mm$

$$K = \frac{b_1/\delta}{5 + b_1/\delta} = 0{,}4 \quad , \quad K \cdot b_1 = 0{,}4 \cdot 10 = 4\,mm$$

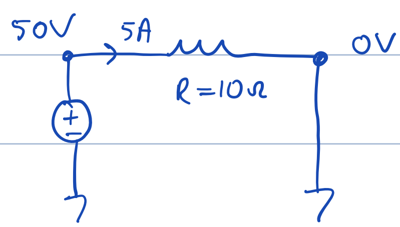
$$l' \approx l - n_v \cdot K b_1 + 2\delta = 990 - 24 \cdot 4 + 2 \cdot 3 = 900\,mm$$

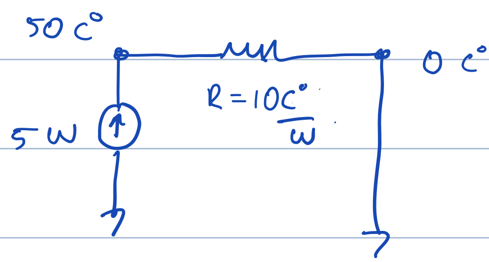


What is the freq. of rotor in synchron machine?

Rotor loss << core loss

Electric Circuit ⟶ Lumped Thermal Network

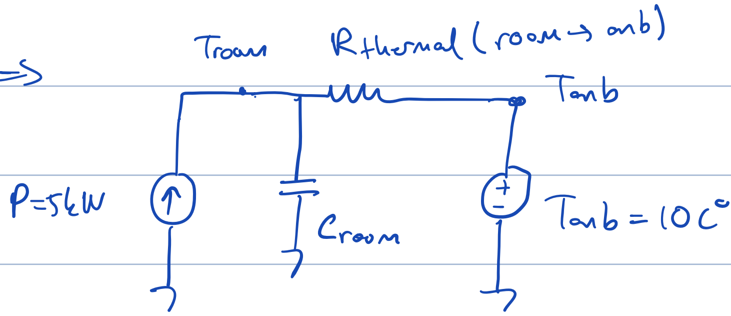




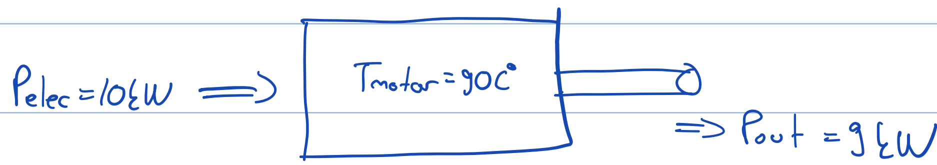
$\Delta W = m.c\,\Delta t$

↳ heat (losses)

⇒



$T_{amb} = 40\ C°$



$\eta = 90\%$

$P_{dissapeted} = 1\,kW$

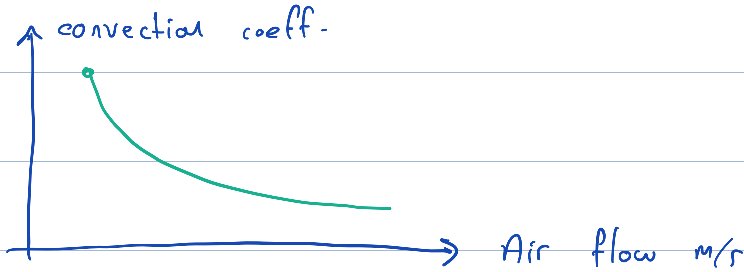
If P_{elec} 50% more s.t. $P_{elec} = 15\,kW$, then $I \nearrow 50\%$

$\Rightarrow P_{loss} = (1{,}5)^2 = 2{,}25\,kW$

$$P = \frac{T_2 - T_1}{R} = \frac{\Delta T}{R} \quad , \quad \Delta T = 50C° \cdot 2{,}25 = 112{,}5\ C°$$

$$T_{motor,new} = 40 + 112{,}5 = 152{,}5\ C°$$

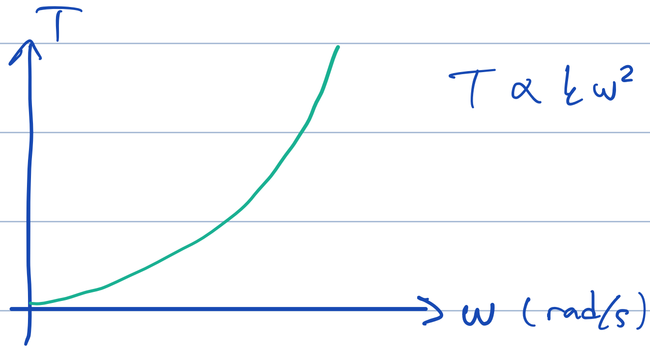
Convection Thermal Resistance:



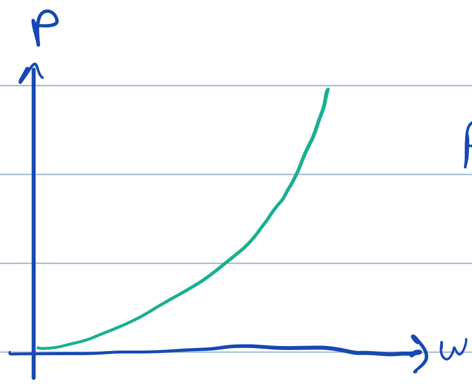
$$R_{conv} = \frac{1}{h \cdot A}$$

A → area
h ↳ convection heat coeff.

Fan Load:



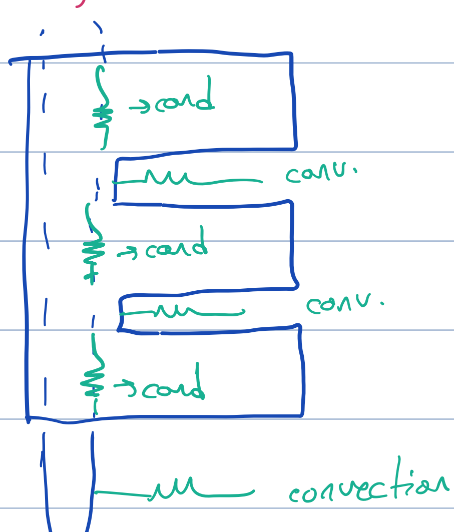
$T \propto k\omega^2$

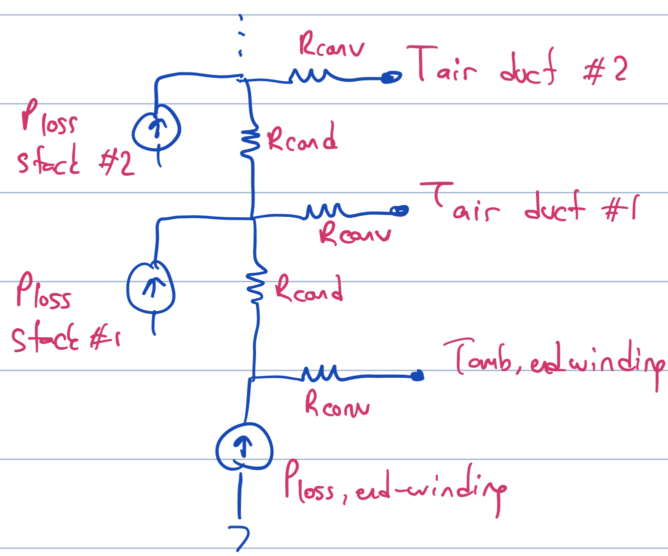


$P = T\omega \propto \omega^3$

If we put a fan on the shaft, and we want to increase the fan speed to double, we need x8 of power.

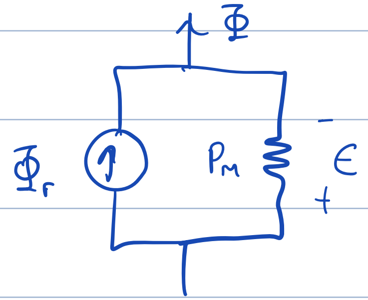
Cooling Duct:





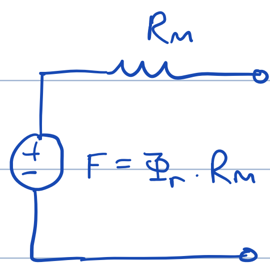
end-winding

PM :

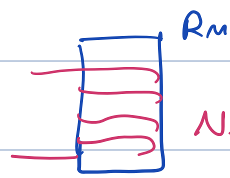


$$P_M = \frac{1}{R}$$

Thevenin Eqv:



$=$

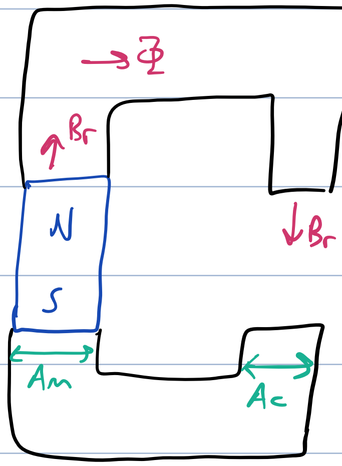


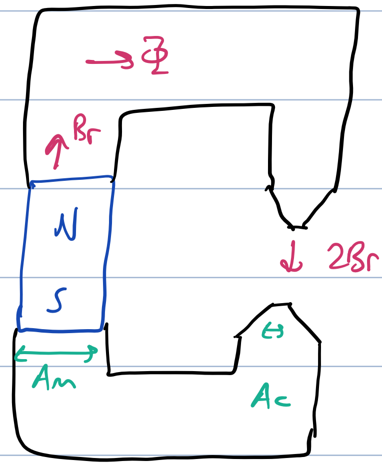
$$N.I = MMF = \frac{B_r \cdot l_m}{\mu_r \cdot \mu_0}$$

If the magnet is operating in linear region

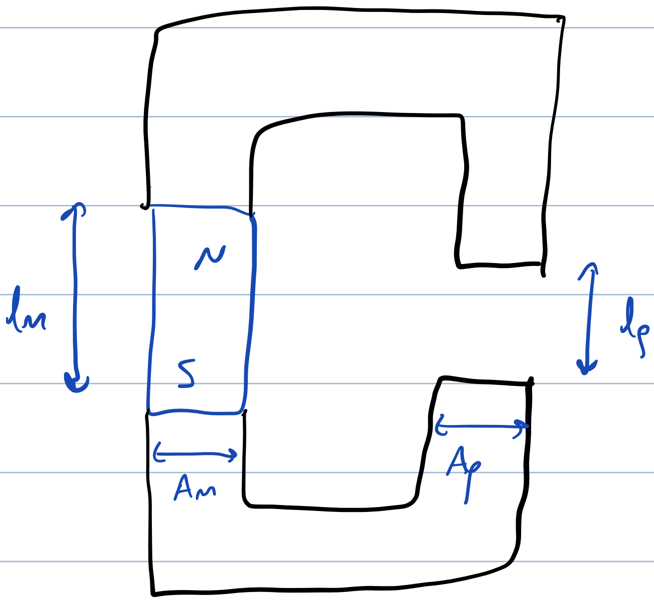
$$B_m = B_r + \mu_r \cdot \mu_0 \cdot H_m, \qquad \mu_r \cong 1,05, \quad H_m \text{ is negative}$$

$$NI = \frac{B_r \cdot l_m}{\mu_r \cdot \mu_0}$$





Ex



Assume $A_g = A_m$

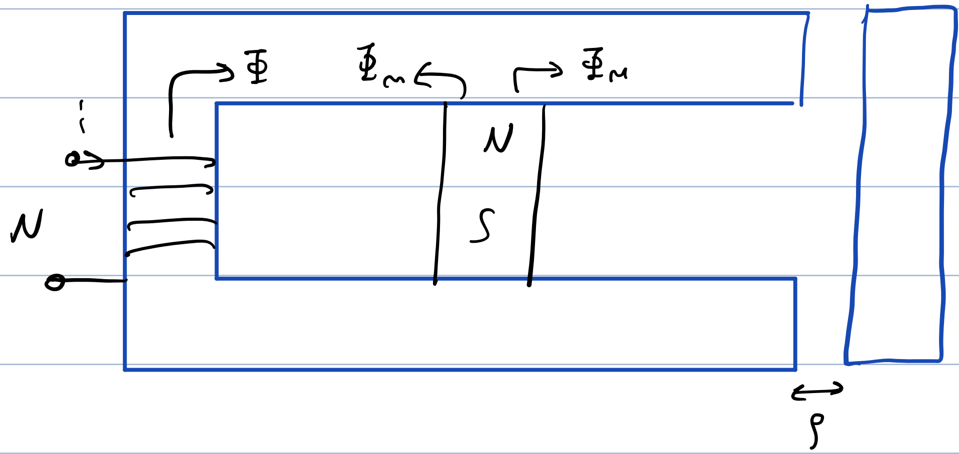
$$NI = \frac{B_r \cdot l_m}{\mu_0 \mu_r} = \Phi \cdot R = \Phi \cdot \left(\frac{l_m + l_g}{\mu_0 A} \right)$$

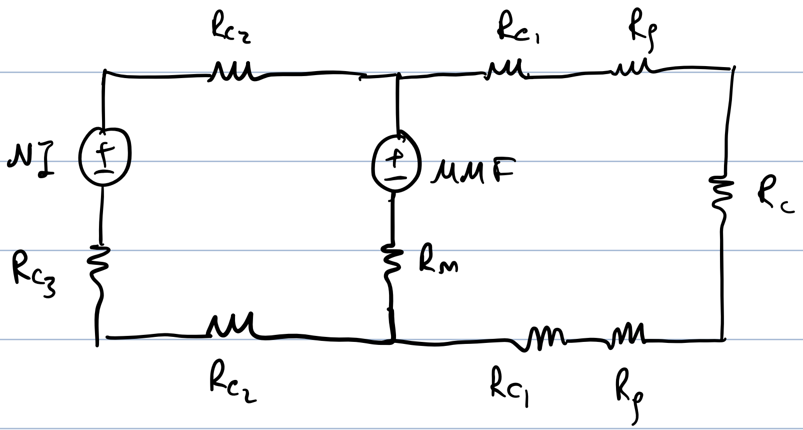
$$NI = \frac{B_r l_m}{\mu_0 \mu_r} \Rightarrow \Phi = \frac{B_r l_m}{\mu_0 \mu_r} \cdot \frac{1}{\frac{l_m + l_g}{\mu_0 A}} = \frac{B_r l_m A}{(l_m + l_g)} \quad (\mu_r \cong 1)$$

$$B_g = \frac{B_r \cdot l_m}{l_m + l_g}$$

if $l_m = l_g$, $B_g = \frac{B_r}{2}$, if you double l_m, $B_g = B_r \frac{2}{3}$

Ex





Ex

If $A_m \neq A_g$

$$\Phi = \frac{B_r\, l_m}{\mu_0} \cdot \frac{1}{\frac{l_m}{\mu_0 A_m} + \frac{l_g}{\mu_0 A_g}} = \frac{B_r\, l_m}{A_g l_m + l_g A_m} \cdot A_g A_m$$

$$B_g = \frac{B_r\, l_m}{l_g + l_m \frac{A_g}{A_m}}$$